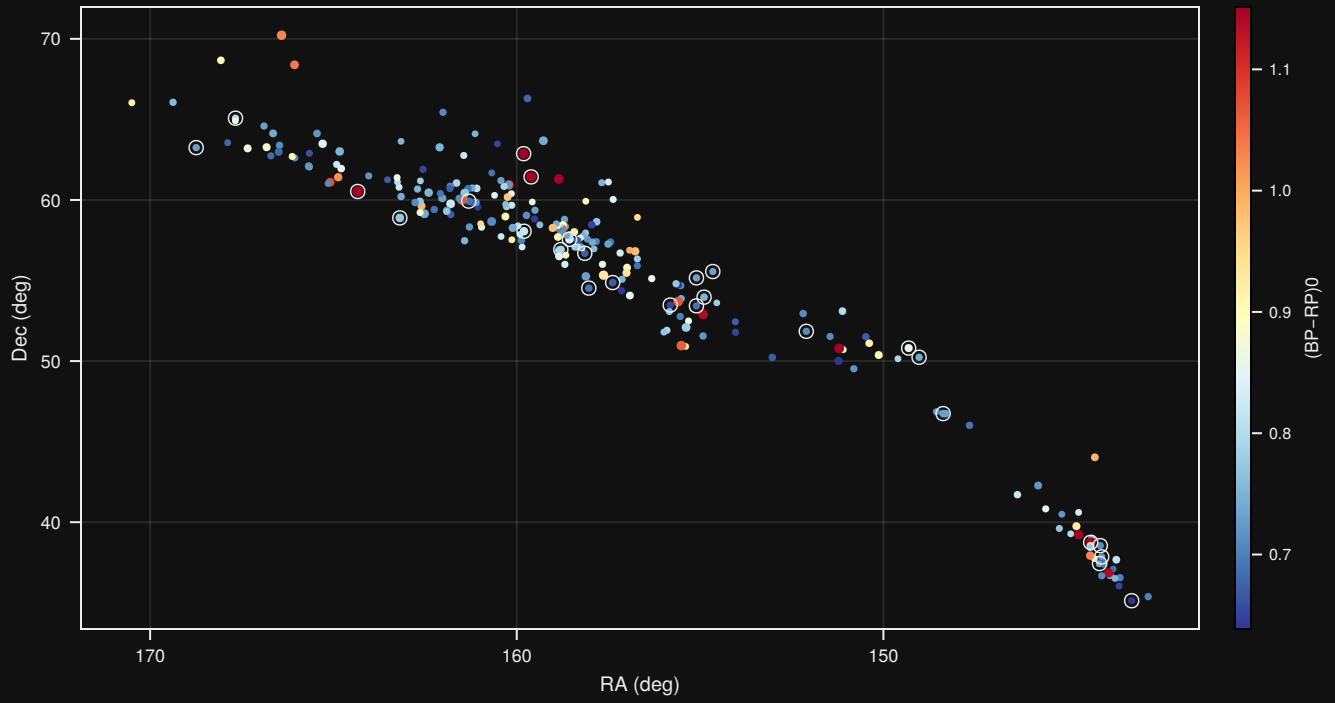
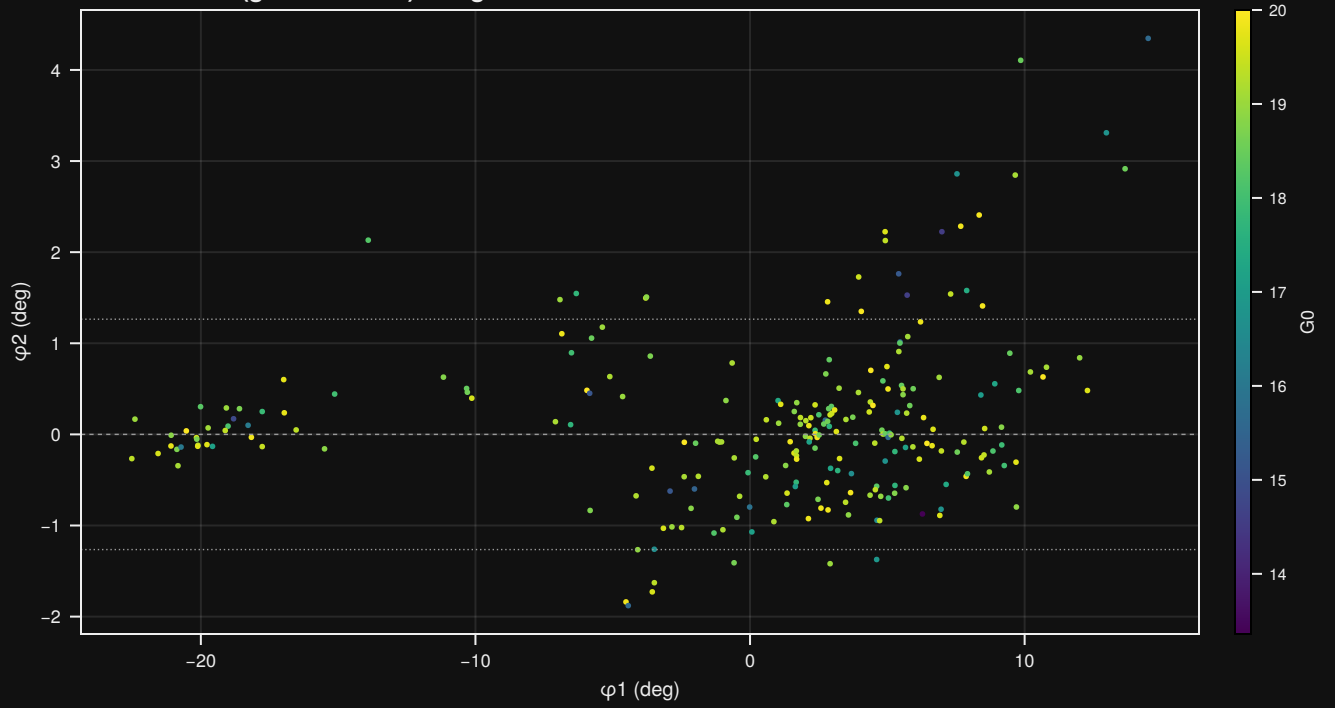
C-24

MEMBERSHIP	
Stars in catalog	244
with Gaia GSP-Phot	115 (47%)
with radial velocity	27
Stream id (sID)	36
GEOMETRY · great-circle fit	
Centroid RA, Dec	157.4, 57.9 deg
Centroid l, b	155.1, 50.7 deg
Pole RA, Dec	41.0, 16.3 deg
Length (φ1, 2–98%)	31.9 deg
Width σ(φ2), robust	0.63 deg
KINEMATICS	
pmRA* med ± σ	0.80 ± 0.77 mas/yr
pmDE med ± σ	-0.43 ± 0.26 mas/yr
μφ1* med ± σ	0.00 ± 0.34 mas/yr
μφ2 med ± σ	-0.85 ± 0.71 mas/yr
Parallax (cat.) med	0.117 ± 0.208 mas
Parallax (wtd, ZP-corr.)	0.139 ± 0.011 mas
Distance	d = 7.2 ± 0.6 kpc
Physical length	4.00 kpc
RV median ± σ(MAD)	-46.4 ± 47.0 km/s
RV gradient dV/dφ1	-1.04 km/s/deg
PHOTOMETRY · POPULATIONS	
G0 range (median)	13.4–20.0 (19.0)
(BP−RP)0 median	0.76
GSP-Phot [M/H] (n=115)	-0.88 ± 0.41
GSP-Phot Teff median	5733 K
RUWE > 1.4	0.9 %
PM clusters (largest)	12 (25 stars)
IBATA+2024 · TABLE 3	
n catalog | paper	244 | 244
n_RV catalog | paper	27 | 27
<[Fe/H]> paper	-0.93
Stellar mass (paper)	2700 Msun
Pole offset vs paper	8.1 deg
Remark	Velocities unclear

Sky map · catalog members (n = 244) · size ∝ brightness · ring = has RV

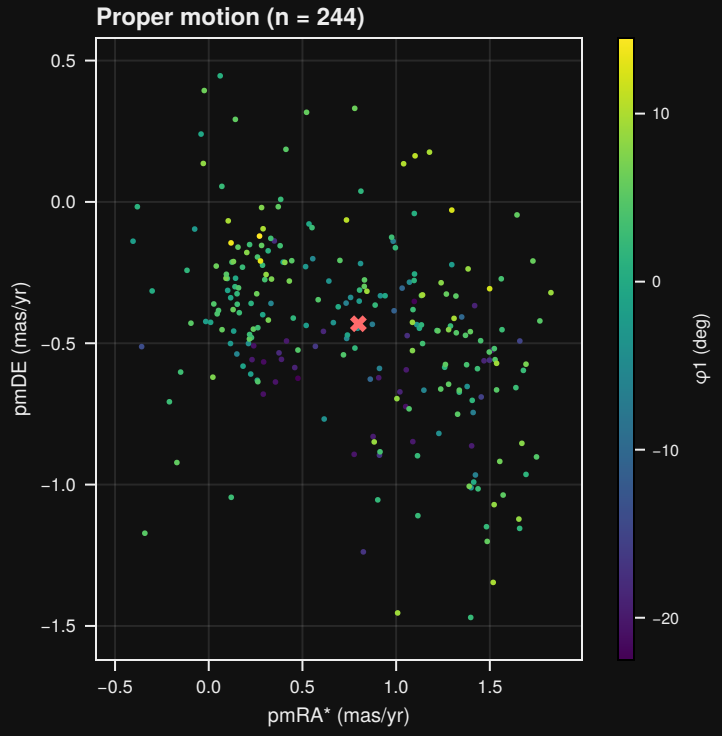


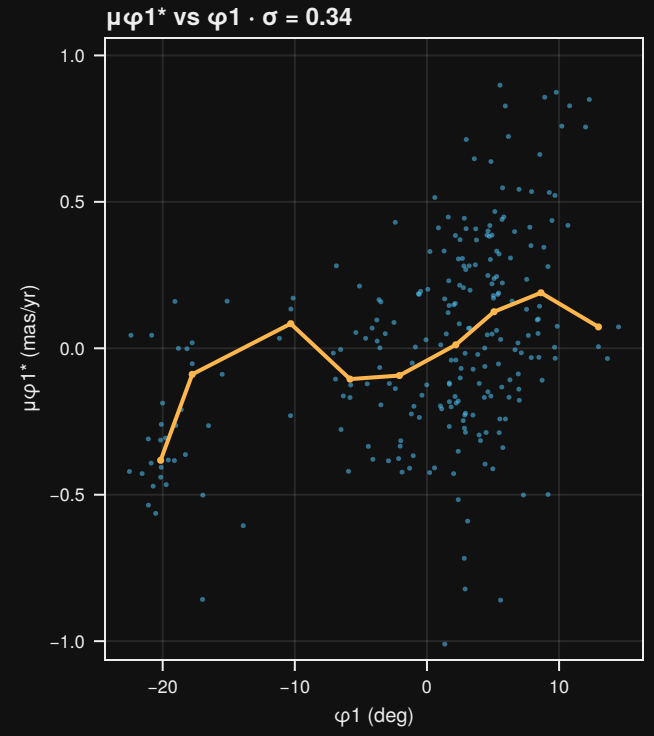
Stream frame (great-circle fit) · length 31.9° · width σ 0.63°

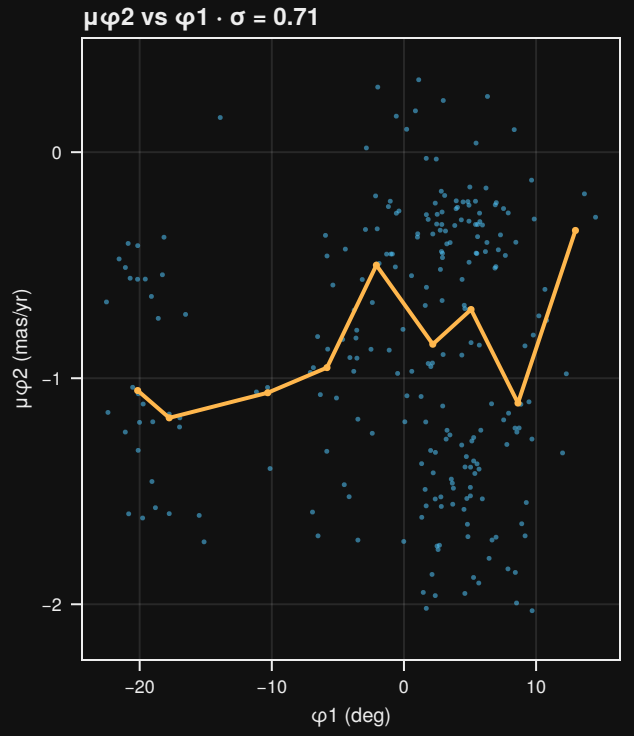


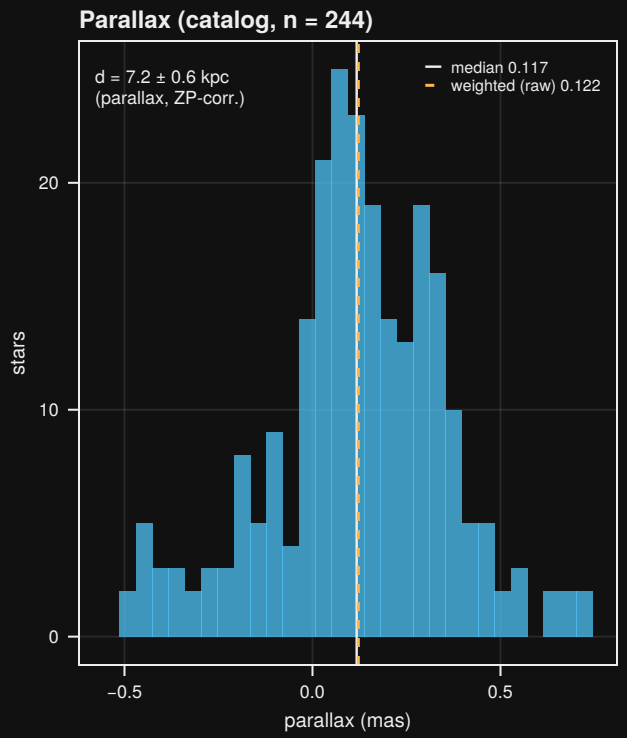
C-24

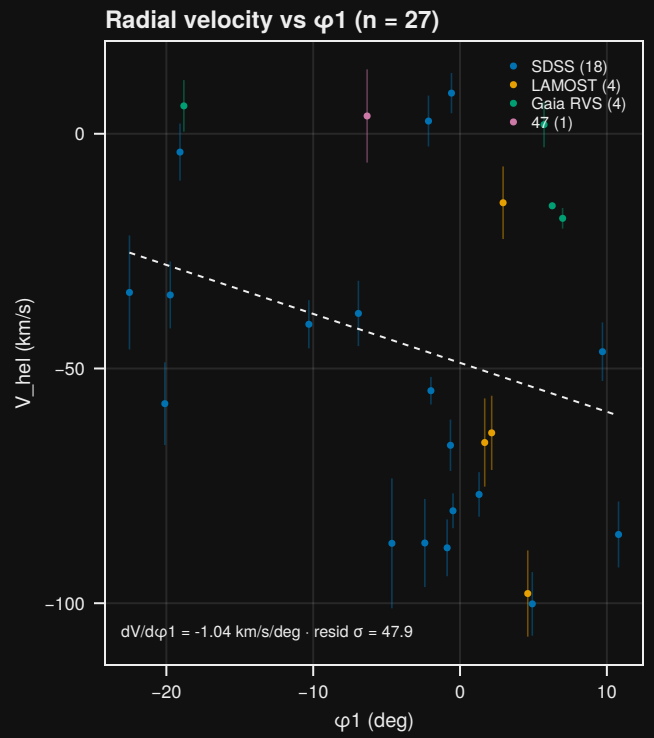
Kinematics · proper motion · parallax · radial velocity

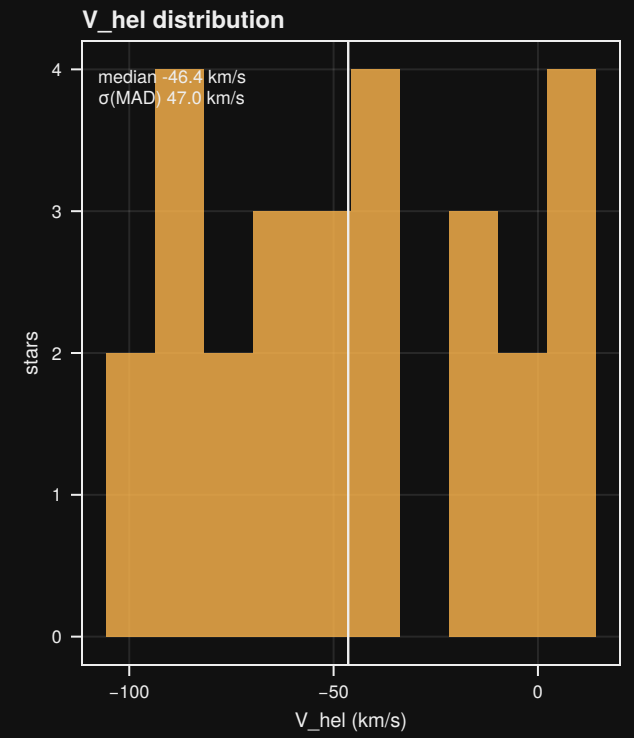






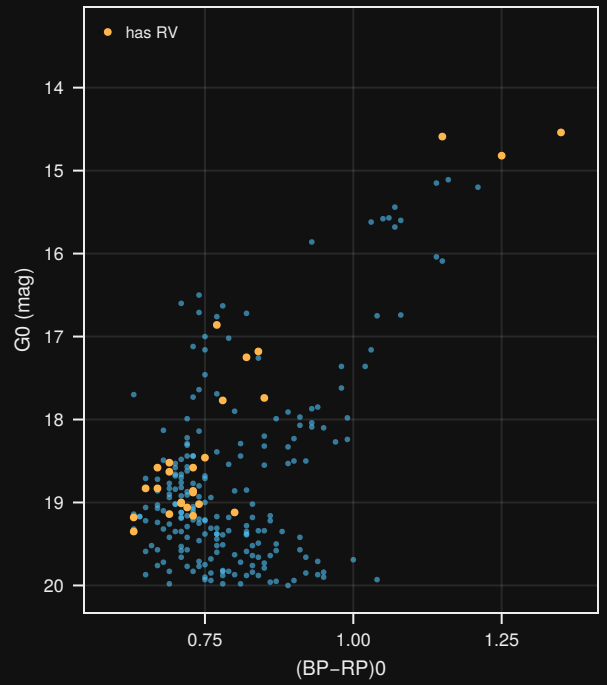




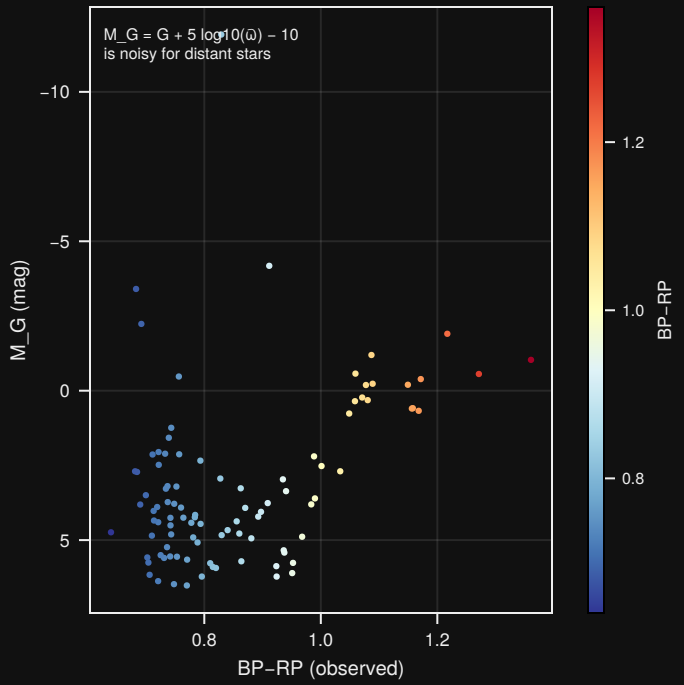


C-24

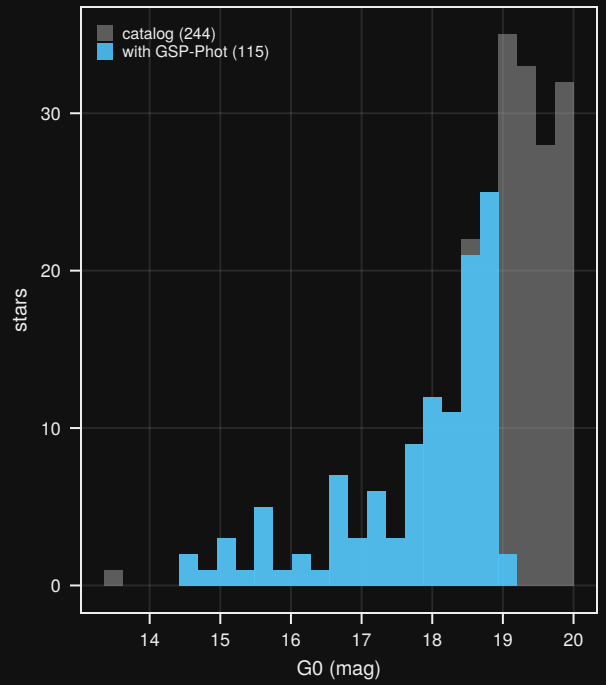
CMD · dereddened (n = 244)



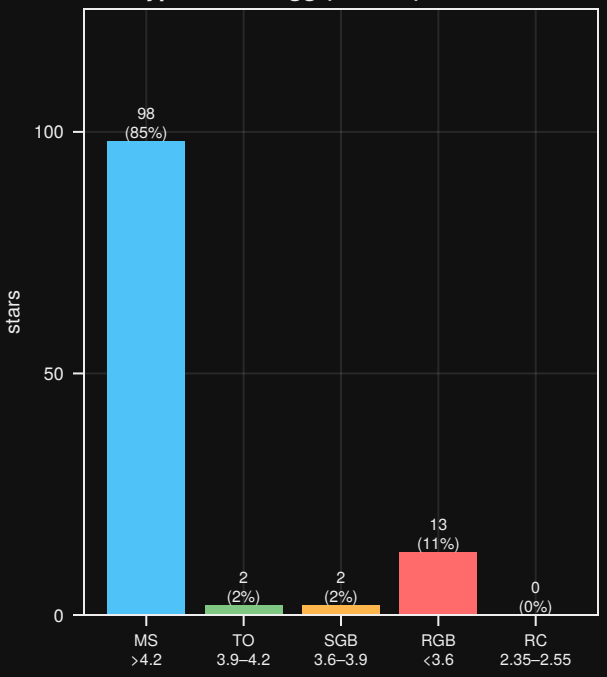
HR diagram · ϖ>0 (n = 97/115)



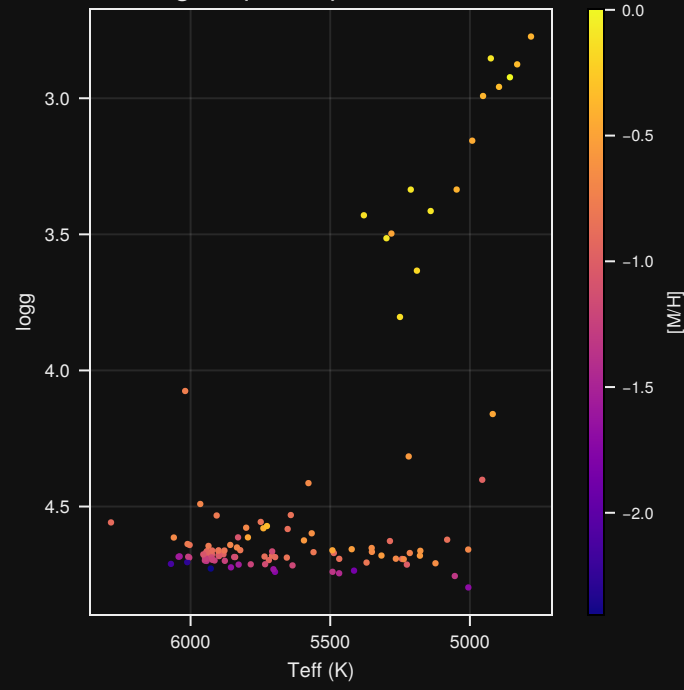
G0 & GSP-Phot coverage



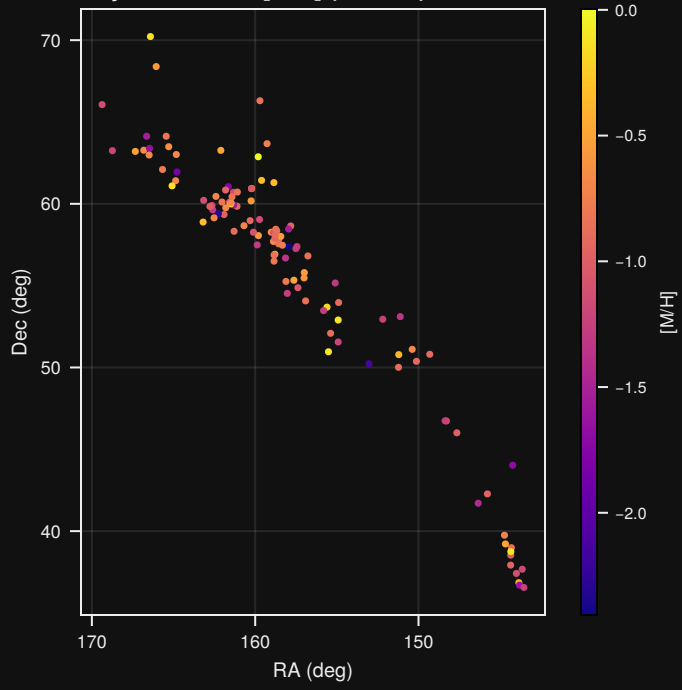
Star types from logg (n = 115)



Kiel diagram (n = 115)



Sky · GSP-Phot [M/H] (n = 115)

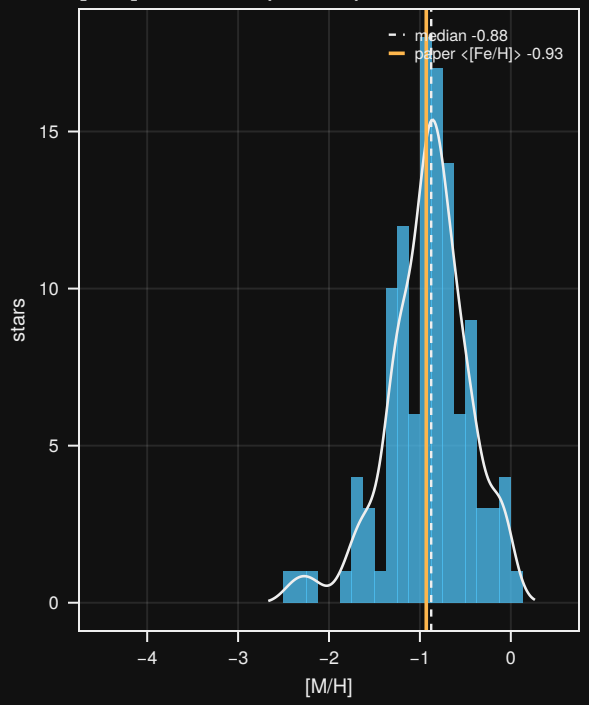


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≳ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

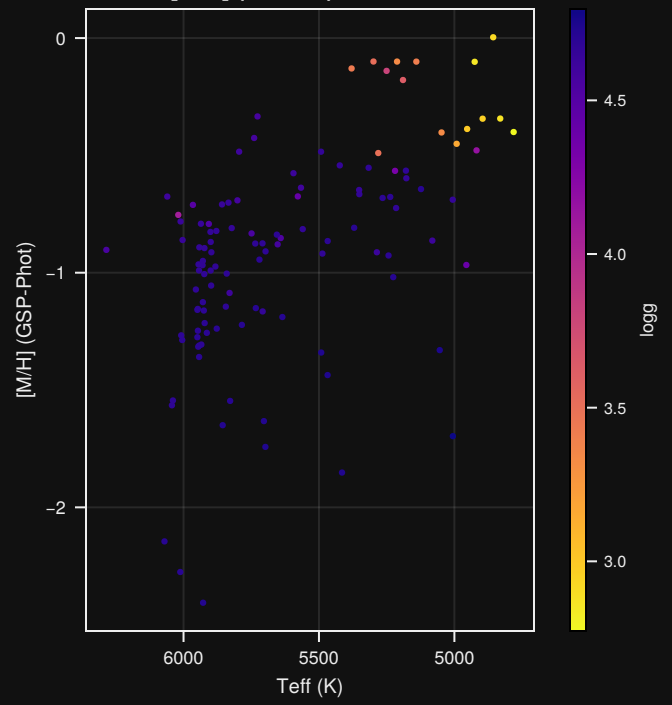
C-24

Metallicity · age · astrometric quality

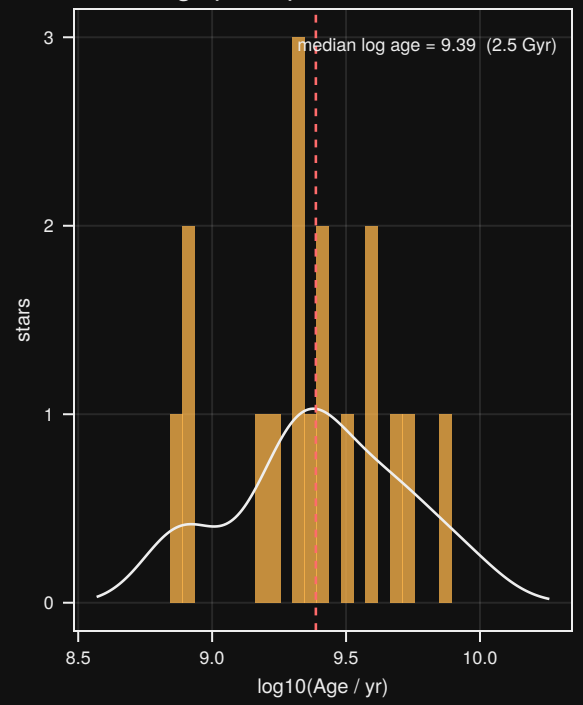
[M/H] GSP-Phot (n = 115)



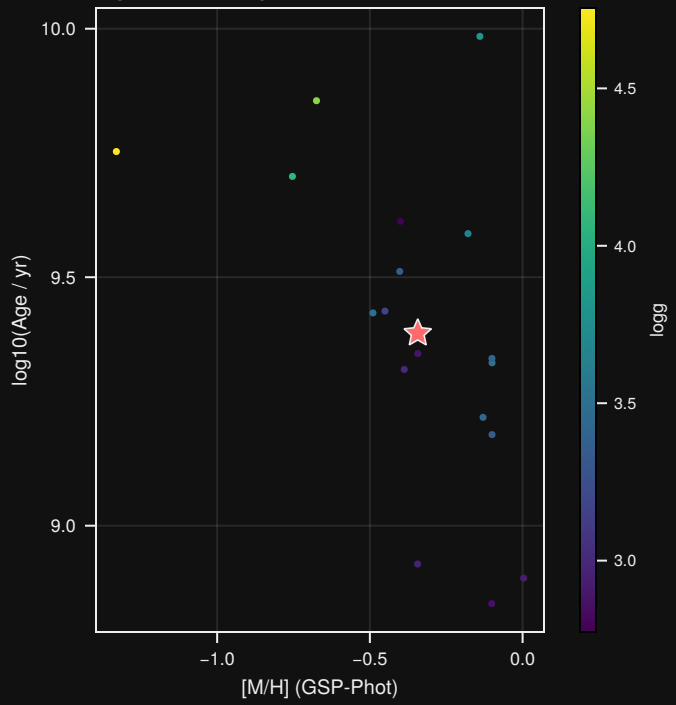
Teff vs [M/H] (n = 115)



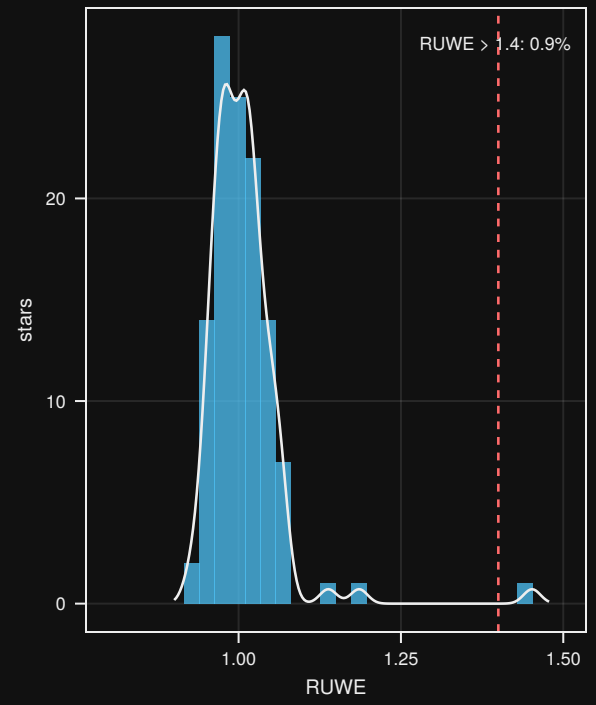
FLAME age (n = 18)



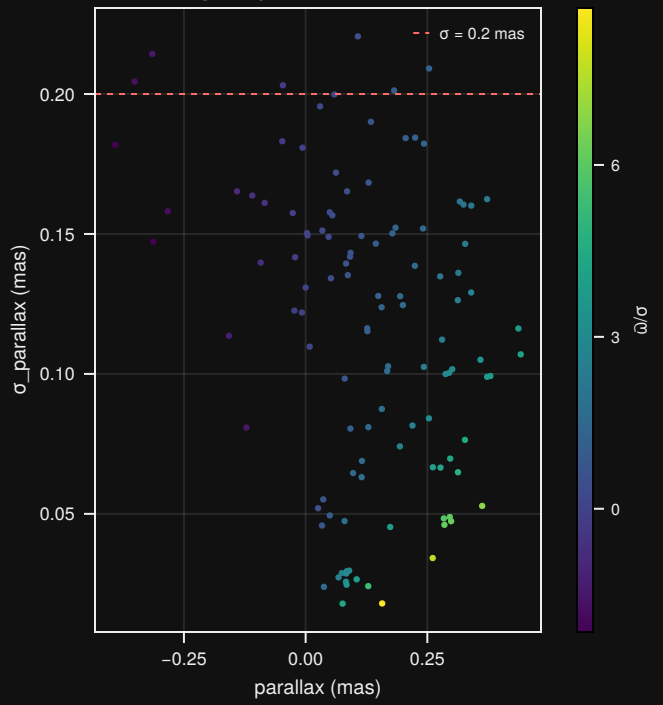
Age–metallicity (n = 18)



RUWE (n = 115)

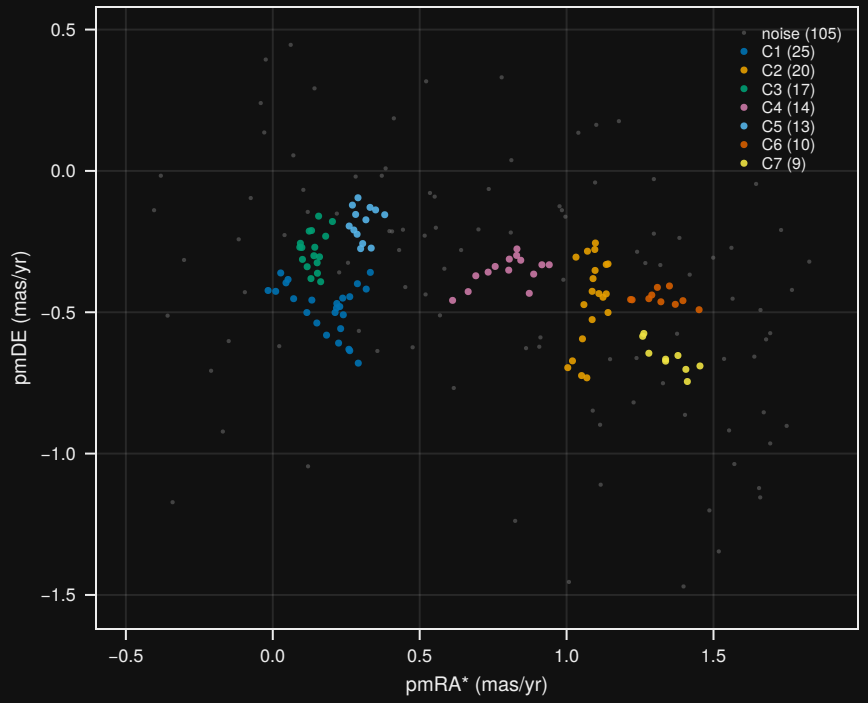


Parallax quality (n = 115)

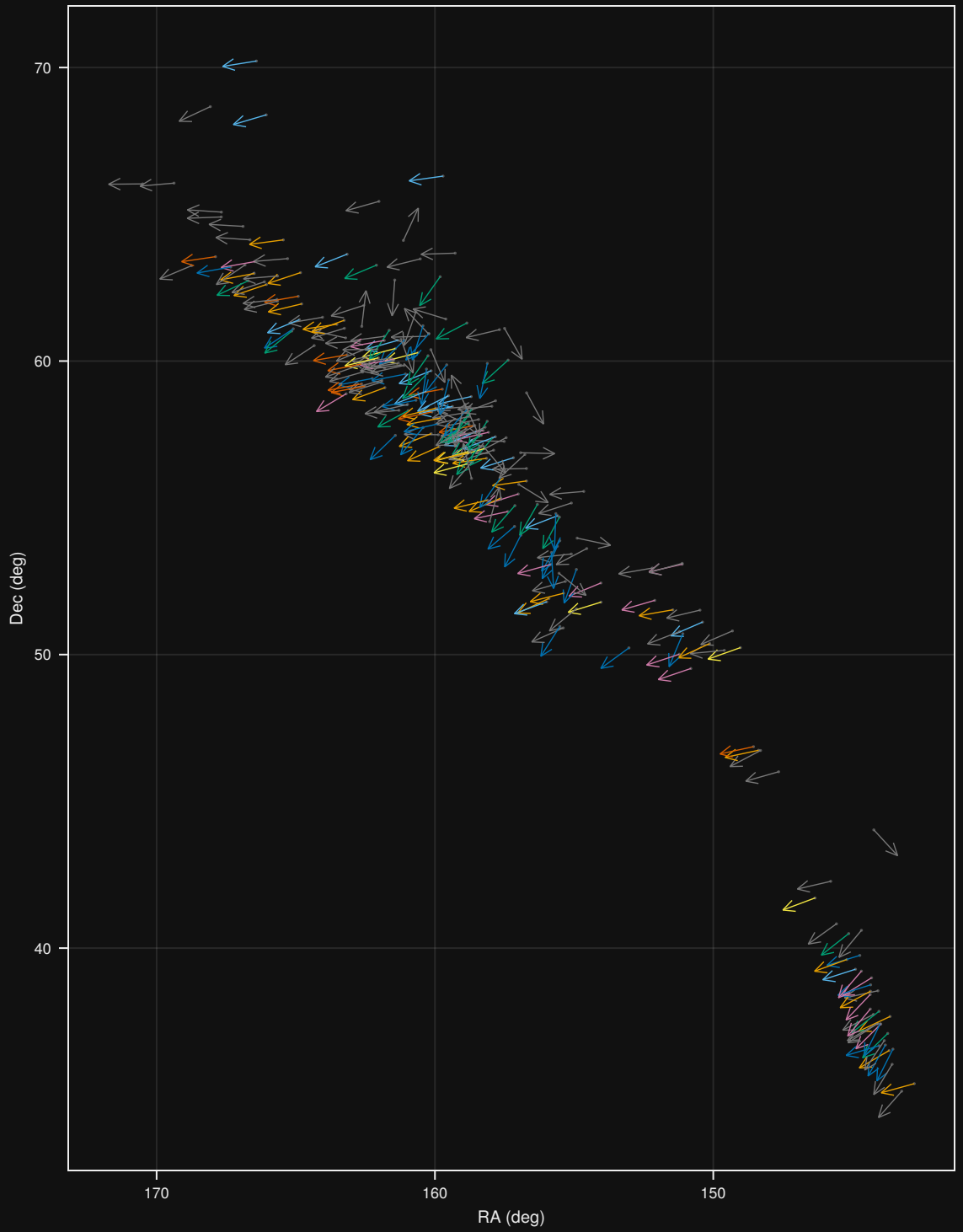


C-24

PM clustering · HDBSCAN · 12 cluster(s) · noise 43%



Proper-motion directions (arrows) coloured by cluster



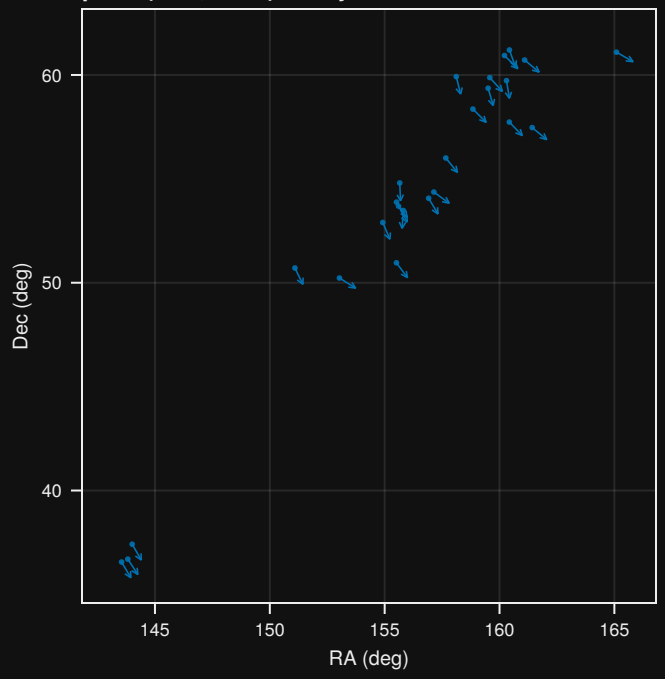
Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	105	0.81	-0.33	0.79	-0.95	15
C1	25	0.22	-0.47	0.10	-0.85	1
C2	20	1.09	-0.43	0.05	-0.85	3
C3	17	0.14	-0.27	0.03	-0.57	1
C4	14	0.82	-0.34	0.10	-0.89	2
C5	13	0.30	-0.17	0.03	-0.77	0
C6	10	1.31	-0.46	0.07	-1.15	0
C7	9	1.34	-0.67	0.10	-0.81	2
C8	8	1.50	-0.56	0.03	-0.71	1

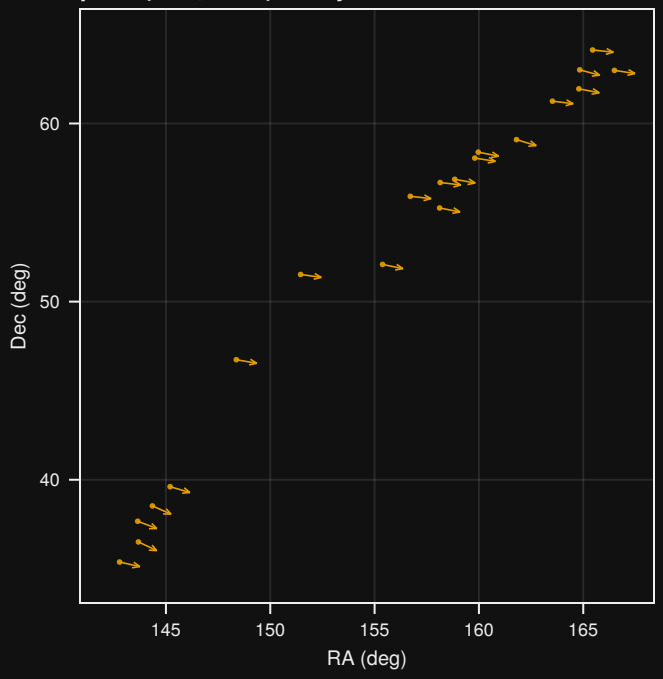
C-24

Per-cluster proper-motion vector maps

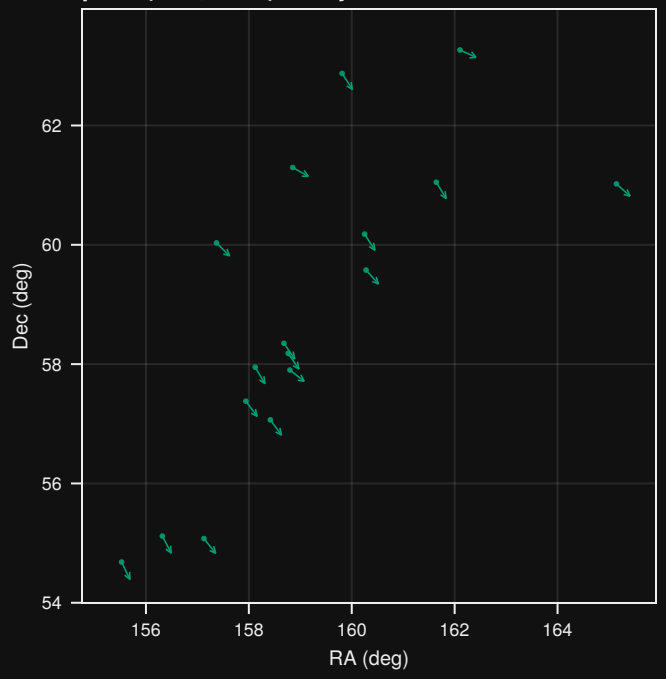
Cluster C1 · n = 25
pm = (0.22, -0.47) mas/yr



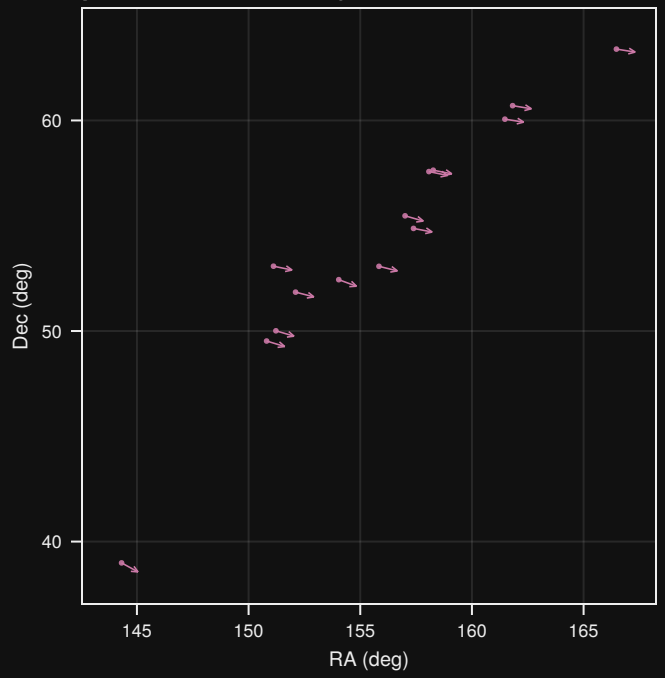
Cluster C2 · n = 20
pm = (1.09, -0.43) mas/yr



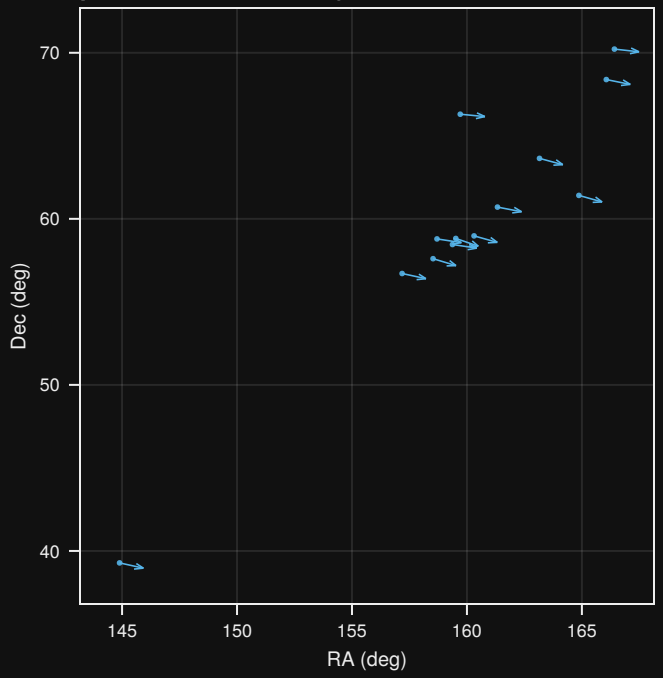
Cluster C3 · n = 17
pm = (0.14, -0.27) mas/yr



Cluster C4 · n = 14
pm = (0.82, -0.34) mas/yr



Cluster C5 · n = 13
pm = (0.30, -0.17) mas/yr



Noise · n = 105

